

Cal 1108.

La integral indefinida.

Motivación: Consideremos la función $f(x) = 2x$

Una función $F_1(x) = x^2$, tiene por derivada a $f(x)$

... $F_2 = x^2 + 1$, también.

Definición de antiderivada:

Sean $f(x)$ y $F(x)$ definidas en el intervalo I , $a, b \in I$

Decimos que $F(x)$ es una antiderivada de $f(x)$ si $F'(x) = f(x)$.

Una de las posibles antiderivadas.

Definición de integral indefinida: Sea $f(x)$ una función, definimos la integral indefinida de $f(x)$ como:

$$\int f(x) dx = F(x) + C$$

donde $F(x)$ es una antiderivada y $C = \mathbb{R}$, llamada constante de integración

Ejemplos: $\int 2x dx = x^2 + C$ • $\int 3 dx = 3x + C$ • $\int \cos(x) dx = \sin(x) + C$

$\int \sec^2(x) dx = \tan(x) + C$ • $\int \sin(x) dx = -\cos(x) + C$

Desde la tabla de derivación tenemos inmediatamente la siguiente tabla de integrales:

• $\int k dx = kx + C$ • $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$

~~$\int \frac{1}{x} dx = \ln|x| + C$~~

$\int x^{-1} dx = \int \frac{1}{x} dx = \ln|x| + C$

• $\int e^x dx = e^x + C$

• $\int \csc^2 x dx = -\cot(x) + C$

• $\int \csc(x) \cot(x) dx = -\csc(x) + C$

• $\int \cos(x) dx = \sin(x) + C$

• $\int -\sin(x) dx = \cos(x) + C$

• $\int \sec^2(x) dx = \tan(x) + C$

• $\int \sec(x) \tan(x) dx = \sec(x) + C$

• $\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin(x) + C$

• $\int \frac{1}{1+x^2} dx = \arctan(x) + C$

$\int \sec(x) dx = \ln|\sec(x) + \tan(x)| + C$

$\int \csc(x) dx = \ln|\csc(x) - \cot(x)| + C$

Propriedades de integral indefinida: sem $f(x)$, $g(x)$ duas funções e $\alpha \in \mathbb{R}$

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

$$\int \alpha f(x) dx = \alpha \int f(x) dx$$

Exemplos. 1) $\int \sqrt{x} dx = \int x^{1/2} dx = \frac{x^{1+1/2}}{1+1/2} + C = \frac{x^{3/2}}{\frac{3}{2}} = \frac{2}{3} \sqrt{x^3} + C$

$$2) \int \left(\frac{x^2 + 3\sqrt{x} + 1}{x} \right) dx = \int \left(x + \frac{3}{\sqrt{x}} + \frac{1}{x} \right) dx = \int x dx + \int 3x^{-1/2} dx + \int \frac{1}{x} dx$$

$$= \frac{x^2}{2} + C + 3 \frac{x^{1/2}}{\frac{1}{2}} + C + \ln|x| + C = \frac{x^2}{2} + 6x^{1/2} + \ln|x| + C$$

$$* \int \frac{5 \sin(x) - 3 \tan(x) - 1}{\sin(x)} dx = \int \frac{5 \sin(x)}{\sin(x)} dx + \int \frac{-3 \tan(x)}{\sin(x)} dx - \int \frac{1}{\sin(x)} dx$$

$$= \int 5 dx - \int 3 \sec(x) dx - \int \csc(x) dx$$

$$= 5x - 3 \ln|\sec(x) + \tan(x)| - \ln|\csc(x) - \cot(x)| + C$$

$$\frac{\tan(x)}{\sin(x)} = \frac{\sin(x)}{\sin(x) \cos(x)}$$

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$$\sin^2(x) + \cos^2(x) = 1 \Rightarrow \frac{\sin^2 x}{\cos^2 x} + \frac{1}{1} = \frac{1}{\cos^2 x} = \tan^2 + 1 = \sec^2(x)$$

$$\tan(x) = \frac{\sin(x)}{\cos(x)}$$

$$\sec(x) \neq \cos^{-1}(x) = (\cos(x))^{-1}$$

$$\csc(x) = (\sin(x))^{-1} \neq \sin^{-1}(x)$$

$$\frac{\sin^2 x}{\sin^2 x} + \frac{\cos^2 x}{\sin^2 x} = \frac{1}{\sin^2(x)} = \cot^2(x) + 1 = \csc^2(x)$$

$$\cot(x) = \frac{\cos(x)}{\sin(x)}$$

$$\int \frac{11 \sec(x) + 4 \cos(x) - x \cos(x) + 7}{\cos(x)} dx$$

$$\int \tan(x) dx = \ln$$

$$= \int (11 \tan(x) + 4 - x + 7 \sec(x)) dx$$

$$= 11 \int \tan(x) dx + \int 4 dx - \int x dx + 7 \int \sec(x) dx$$

$$= -11 (\ln |\cos(x)|) + 4x - \frac{x^2}{2} + 7 \ln |\sec(x) + \tan(x)| + C$$

$$\int \left(\frac{5e^x - 3}{e^x} \right) dx = \int 5 dx - \int \frac{3}{e^x} dx$$

$$= 5x - \int 3e^{-x} dx = 5x - 3 \int e^{-x} dx$$

$$= 5x + 3e^{-x}$$

Exercises.

$$c) \int (x^4 - 2x^2) dx$$

$$= \int x^4 dx - 2 \int x^2 dx$$

$$= \frac{x^5}{5} - \frac{2}{3} x^3$$

$$d) \int \left(\frac{2}{x^3} + 4x^2 \right) dx = \int \frac{2}{x^3} dx + \int 4x^2 dx$$

$$= \int 2x^{-3} dx + \frac{4}{3} x^3$$

$$= -x^{-2} + \frac{4}{3} x^3$$

$$1) \int (3 \sec^2(x) - 4 \tan(x)) dx = 3 \int \sec^2(x) dx - 4 \int \tan(x) dx = 3 \tan(x) - 4 \ln |\cos(x)| + C$$

$$= 3 \tan(x) - \ln |\cos(x)| + C$$